EXERCISE 1 Without using a calculator, find :

1. $\csc \frac{\pi}{3}$
2. $\cot \frac{2\pi}{3}$
3. $\sec \frac{5\pi}{6}$
4. $\cot \pi$

EXERCISE 2 Without using a calculator, find $\csc x$, $\sec x$, and $\cot x$ given :

1. $\sin x = \frac{3}{5}, \quad 0 \leq x \leq \frac{\pi}{2}$
2. $\cos x = \frac{2}{3}, \quad \frac{3\pi}{2} < x < 2\pi$

EXERCISE 3 Find the other *five trigonometric ratios* if :

1. $\cos x = \frac{3}{4}$ and $\frac{3\pi}{2} < x < 2\pi$
2. $\sin x = -\frac{2}{3}$ and $\pi < x < \frac{3\pi}{2}$
3. $\sec x = 2\frac{1}{2}$ and $0 < x < \frac{\pi}{2}$
4. $\csc x = 2$ and $\frac{\pi}{2} < x < \pi$
5. $\tan \beta = \frac{1}{2}$ and $\pi < \beta < \frac{3\pi}{2}$
6. $\cot \theta = \frac{4}{3}$ and $\pi < \theta < \frac{3\pi}{2}$

EXERCISE 4 Using the graph of $y = \cos x$, sketch the graph of

$$y = \frac{1}{\cos x} = \sec x$$

for $-2\pi \leq x \leq 2\pi$. Check your answer using technology.

EXERCISE 5 Using the graph of $y = \tan x$, sketch the graph of

$$y = \frac{1}{\tan x} = \cot x$$

for $-2\pi \leq x \leq 2\pi$.

EXERCISE 6 Use technology to sketch $y = \sec x$ and $y = \csc(x + \frac{\pi}{2})$ on the same set of axes for $-2\pi \leq x \leq 2\pi$. Explain your answer.

EXERCISE 7 Solve exactly for $0 \leq x \leq 2\pi$:

1. $\sec x = 2$
2. $\csc x = -\sqrt{2}$
3. $\sqrt{3} \sec 2x = -2$
4. $\csc(x + \frac{\pi}{6}) + \sqrt{2} = 0$
5. $\cot x + 1 = 0$
6. $\cot(2x - \frac{\pi}{4}) - \sqrt{3} = 0$

EXERCISE 8 Find, giving your answer in radians :

1. $\arccos 1$
2. $\arcsin(-1)$
3. $\arctan 1$
4. $\arctan(-1)$
5. $\arcsin \frac{1}{2}$
6. $\arccos\left(-\frac{\sqrt{3}}{2}\right)$
7. $\arctan \sqrt{3}$
8. $\arccos\left(-\frac{1}{\sqrt{2}}\right)$
9. $\arctan\left(-\frac{1}{\sqrt{3}}\right)$
10. $\sin^{-1}(-0.767)$
11. $\cos^{-1} 0.327$
12. $\tan^{-1}(-50)$

EXERCISE 9 Find the invariant point for the inverse transformation from :

1. $y = \sin x$ to $y = \arcsin x$
2. $y = \tan x$ to $y = \arctan x$
3. $y = \cos x$ to $y = \arccos x$

EXERCISE 10 Simplify :

1. $\arcsin(\sin \frac{\pi}{3})$
2. $\arccos(\cos(-\frac{\pi}{6}))$
3. $\tan(\arctan(0.3))$
4. $\cos(\arccos(-\frac{1}{2}))$
5. $\arctan(\tan \pi)$
6. $\arcsin(\sin \frac{4\pi}{3})$

EXERCISE 11 Simplify :

1. $\sin\theta + \sin\theta$
2. $2\cos\theta + \cos\theta$
3. $3\sin\theta - \sin\theta$
4. $3\sin\theta - 2\sin\theta$
5. $\tan\theta - 3\tan\theta$
6. $2\cos^2\theta - 5\cos^2\theta$

EXERCISE 12 Simplify :

1. $3\tan x - \dfrac{\sin x}{\cos x}$
2. $\dfrac{\sin^2 x}{\cos^2 x}$
3. $\tan x \cos x$
4. $\dfrac{\sin x}{\tan x}$
5. $3\sin x + 2\cos x \tan x$
6. $\dfrac{2\tan x}{\sin x}$

EXERCISE 13 Simplify :

1. $\tan x \cot x$
2. $\sin x \csc x$
3. $\csc x \cot x$
4. $\sin x \cot x$
5. $\dfrac{\cot x}{\csc x}$

EXERCISE 14 Simplify :

1. $3\cos\theta - \cos(-\theta)$
2. $\tan(-\theta)$
3. $\sin(-\theta) + \cos\left(\frac{\pi}{2} - \theta\right)$
4. $\tan(\pi - \theta)$
5. $\tan\left(\frac{\pi}{2} - \theta\right)$
6. $\sin\left(\frac{\pi}{2} - \theta\right) - \cos(\pi - \theta)$

7. $\dfrac{\sin(-\theta)}{\cos(\pi-\theta)}$
8. $\dfrac{\cos\left(\frac{\pi}{2}-\theta\right)}{\cos(-\theta)}$
9. $\dfrac{\sin(\pi-\theta)-\sin(-\theta)}{\cos(-\theta)}$

EXERCISE 15 Simplify :

1. $3\sin^2\theta+3\cos^2\theta$
2. $-2\sin^2\theta-2\cos^2\theta$
3. $-\cos^2\theta-\sin^2\theta$
4. $3-3\sin^2\theta$
5. $4-4\cos^2\theta$
6. $\cos^3\theta+\cos\theta\sin^2\theta$
7. $\cos^2\theta-1$
8. $\sin^2\theta-1$
9. $2\cos^2\theta-2$
10. $\dfrac{1-\sin^2\theta}{\cos^2\theta}$
11. $\dfrac{1-\cos^2\theta}{\sin\theta}$
12. $\dfrac{\cos^2\theta-1}{-\sin\theta}$

EXERCISE 16 Prove that $1+\cot^2\theta=\csc^2\theta$.

EXERCISE 17 Expand and simplify, if possible :

1. $(1+\sin\theta)^2$
2. $(\sin\alpha-2)^2$
3. $(\tan\alpha-1)^2$
4. $(\sin\alpha+\cos\alpha)^2$
5. $(\sin\beta-\cos\beta)^2$
6. $-(2-\cos\alpha)^2$

EXERCISE 18 Simplify :

1. $1-\sec^2\beta$

2. $\dfrac{\tan^2\theta(\cot^2\theta+1)}{\tan^2\theta+1}$
3. $\cos^2\alpha(\sec^2\alpha-1)$
4. $(\sin x+\tan x)(\sin x-\tan x)$
5. $(2\sin\theta+3\cos\theta)^2+(3\sin\theta-2\cos\theta)^2$
6. $(1+\csc\theta)(\sin\theta-\sin^2\theta)$
7. $\sec A-\sin A\tan A-\cos A$

EXERCISE 19 Factorise :

1. $1-\sin^2\theta$
2. $\sin^2\alpha-\cos^2\alpha$
3. $\tan^2\alpha-1$
4. $2\sin^2\beta-\sin\beta$
5. $2\cos\phi+3\cos^2\phi$
6. $3\sin^2\theta-6\sin\theta$
7. $\tan^2\theta+5\tan\theta+6$
8. $2\cos^2\theta+7\cos\theta+3$
9. $6\cos^2\alpha-\cos\alpha-1$
10. $3\tan^2\alpha-2\tan\alpha$
11. $\sec^2\beta-\csc^2\beta$
12. $2\cot^2 x-3\cot x+1$
13. $2\sin^2 x+7\sin x\cos x+3\cos^2 x$

EXERCISE 20 Simplify :

1. $\dfrac{1-\sin^2\alpha}{1-\sin\alpha}$
2. $\dfrac{\tan^2\beta-1}{\tan\beta+1}$
3. $\dfrac{\cos^2\phi-\sin^2\phi}{\cos\phi+\sin\phi}$
4. $\dfrac{\cos^2\phi-\sin^2\phi}{\cos\phi-\sin\phi}$
5. $\dfrac{\sin\alpha+\cos\alpha}{\sin^2\alpha-\cos^2\alpha}$

6. $\dfrac{3-3\sin^2\theta}{6\cos\theta}$
7. $1-\dfrac{\cos^2\theta}{1+\sin\theta}$
8. $\dfrac{1+\cot\theta}{\csc\theta}-\dfrac{\sec\theta}{\tan\theta+\cot\theta}$
9. $\dfrac{\tan^2\theta}{\sec\theta-1}$

EXERCISE 21 Show that :

1. $(\cos\theta+\sin\theta)^2+(\cos\theta-\sin\theta)^2=2$
2. $(\sin\theta+4\cos\theta)^2+(4\sin\theta-\cos\theta)^2=17$
3. $(1-\cos\theta)\left(1+\dfrac{1}{\cos\theta}\right)=\tan\theta\sin\theta$
4. $\left(1+\dfrac{1}{\sin\theta}\right)(\sin\theta-\sin^2\theta)=\cos^2\theta$

EXERCISE 22 Solve for $0\le x\le 2\pi$:

1. $2\cos^2 x=\sin x+1$
2. $\sin^2 x=2-\cos x$
3. $2\cos^2 x=3\sin x$
4. $2\tan^2 x+3\sec^2 x=7$

EXERCISE 23 For $-\pi\le x\le\pi$, find the exact solutions of

$$3\sec 2x=\cot 2x+3\tan 2x.$$

EXERCISE 24

For $\theta=30°$, verify that:

a $\sin 2\theta=2\sin\theta\cos\theta$ b $\cos 2\theta=\cos^2\theta-\sin^2\theta$ c $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}$

EXERCISE 25

If $\sin\theta=\frac{4}{5}$ and $\cos\theta=\frac{3}{5}$ find the exact value of:

a $\sin 2\theta$ b $\cos 2\theta$ c $\tan 2\theta$

EXERCISE 26

If α is acute and $\cos 2\alpha=-\frac{7}{9}$, find without a calculator:

a $\cos\alpha$ b $\sin\alpha$

EXERCISE 27

If α is acute and $\cos 2\alpha = -\frac{7}{9}$, find without a calculator:

a $\cos\alpha$ **b** $\sin\alpha$

EXERCISE 28

Find the exact value of $\tan A$ if:

a $\tan 2A = \frac{21}{20}$ and A is obtuse **b** $\tan 2A = -\frac{12}{5}$ and A is acute.

EXERCISE 29

Use an appropriate double angle identity to simplify:

a $2\sin\alpha\cos\alpha$ **b** $4\cos\alpha\sin\alpha$ **c** $\sin\alpha\cos\alpha$

d $2\cos^2\beta - 1$ **e** $1 - 2\cos^2\phi$ **f** $1 - 2\sin^2 N$

g $2\sin^2 M - 1$ **h** $\cos^2\alpha - \sin^2\alpha$ **i** $\sin^2\alpha - \cos^2\alpha$

j $2\sin 2A\cos 2A$ **k** $2\cos 3\alpha\sin 3\alpha$ **l** $2\cos^2 4\theta - 1$

m $1 - 2\cos^2 3\beta$ **n** $1 - 2\sin^2 5\alpha$ **o** $2\sin^2 3D - 1$

p $\cos^2 2A - \sin^2 2A$ **q** $\cos^2\left(\frac{\alpha}{2}\right) - \sin^2\left(\frac{\alpha}{2}\right)$ **r** $2\sin^2 3P - 2\cos^2 3P$

EXERCISE 30

Find the exact value of $\left[\cos\frac{\pi}{12} + \sin\frac{\pi}{12}\right]^2$.

EXERCISE 31

Show that:

a $(\sin\theta + \cos\theta)^2 = 1 + \sin 2\theta$ **b** $\cos^4\theta - \sin^4\theta = \cos 2\theta$

Check these identities by graphing both sides of the equations on the same set of axes.

EXERCISE 32

Solve exactly for x where $0 \leqslant x \leqslant 2\pi$:

a $\sin 2x + \sin x = 0$ **b** $\sin 2x - 2\cos x = 0$ **c** $\sin 2x + 3\sin x = 0$

EXERCISE 33

Prove that:

a $\dfrac{\sin 2\theta}{1-\cos 2\theta} = \cot\theta$ **b** $\dfrac{\sin\theta + \sin 2\theta}{1+\cos\theta+\cos 2\theta} = \tan\theta$ **c** $\dfrac{\sin 2\theta}{1+\cos 2\theta} = \tan\theta$

d $\operatorname{cosec} 2\theta = \tan\theta + \cot 2\theta$ **e** $\dfrac{\sin 2\theta}{\sin\theta} - \dfrac{\cos 2\theta}{\cos\theta} = \sec\theta$